

Part 6 - Business Flow and Interview Strategy

A study guide for explaining the project end-to-end and prioritizing the 48-hour build.

How to read this pack

Blue boxes are explicit case-study requirements recovered from the source explanation. Green boxes are implementation recommendations or inferences. This distinction helps you explain what was mandatory and what you chose as a developer.

Language style: simple Hinglish, interview-focused, and implementation-oriented. The goal is to help you build the project and confidently explain why each part exists.

1. Complete Business Flow

Interview mein agar ek hi flow perfectly explain karna ho, to customer -> challan -> confirm -> stock reduction -> stock movement log explain karo. Ye application ka heart hai.

ABC Traders wants:
10 Monitors
5 Keyboards

Sales logs in -> searches customer -> creates draft challan
Draft challan -> no stock reduction
Sales confirms -> backend checks stock
Enough stock -> reduce stock -> create OUT movements -> mark confirmed
Not enough stock -> rollback -> return insufficient stock error

2. End-to-End Explanation In Hinglish

Ye project ek internal Mini ERP + CRM portal hai jo wholesale/distribution company ke daily operations ko manage karta hai. Admin users aur overall data manage karta hai. Sales team customers, follow-ups aur sales challans create karti hai. Warehouse team products and stock movements track karti hai. Accounts team confirmed sales data dekh sakti hai.

Jab Sales user challan banata hai, system pehle usse Draft status mein save karta hai. Draft stage par stock reduce nahi hota because order final nahi hua. Jab user Confirm karta hai, backend database transaction ke andar har product ka stock check karta hai. Agar stock enough hai, stock reduce hota hai and stock movement log mein OUT entry create hoti hai. Agar stock insufficient hai, operation reject hota hai and stock negative nahi hota.

Challan items product snapshot data store karte hain, jaise product name, SKU and unit price. Isse old challans correct rahte hain even if product price baad mein change ho jaye. Ye real-world accounting and audit requirement ke liye important hai.

3. What They Are Evaluating

Area	What they want to see
Frontend	Usable React admin UI, forms, tables, responsiveness
Backend	Clean REST APIs, validation, error handling, status codes
Database	Correct relationships and realistic data modeling
Business logic	Draft vs confirmed, stock reduction, no negative stock
Engineering practices	JWT, roles, transactions, environment variables, Git commits
Deployment	Working live app and clear README

4. 48-Hour Build Strategy

Time	Work
0-2 hr	Architecture, screens list, database schema and repo setup
2-6 hr	Backend setup, database connection, ORM/migrations
6-10 hr	Authentication, JWT, roles and seed users
10-14 hr	Customer CRM CRUD, search and notes
14-18 hr	Products, inventory fields, low-stock display

Time	Work
18-24 hr	Challan draft creation, items and snapshot fields
24-28 hr	Confirm challan transaction, stock validation and movement log
28-34 hr	React screens and API integration
34-38 hr	Role-based UI, loading/error states, responsive cleanup
38-42 hr	Deployment frontend/backend/database
42-46 hr	README, API docs, test credentials, seed data
46-48 hr	Final testing, demo script and bug fixes

Priority warning

Bonus features like Docker, GitHub Actions, PDF invoice or S3 uploads should come only after core auth, CRM, inventory, challan, stock validation, deployment and README are complete.

5. Must-Have vs Bonus

Must-have	Bonus after core is done
Auth + roles	Docker
Customer CRM + follow-up notes	GitHub Actions
Products + inventory + low stock	PDF invoice/challan export
Stock movement audit trail	AWS S3 product images
Draft/confirmed/cancelled challans	Advanced analytics
No negative stock validation	Email/SMS reminders
Deployment + README	Advanced AWS infrastructure

6. Interview Questions and Answers

Question	Strong answer
Why JWT?	It allows stateless authentication. Backend verifies token and role on every protected request.
Why stock validation in backend?	Frontend can be bypassed. Backend/database transaction is the trusted enforcement layer.
Why transaction during confirmation?	Stock reduction, movement log and challan status must succeed or fail together.
Why product snapshot?	Old challans should not change when product name or price changes later.
Why movement log?	It provides audit trail: what changed, why, by whom and when.
Why Draft status?	Sales can prepare challan without reducing stock before final confirmation.
What is 409 Conflict?	A business rule conflict, for example requested stock is more than available stock.

7. Short Presentation Script

Reusable interview explanation

I built this as a Mini ERP + CRM portal for a distribution business. The CRM side manages customers, follow-ups and notes. The ERP side manages products, inventory, stock movements and sales challans. Authentication uses JWT and role-based access for Admin, Sales, Warehouse and Accounts. The main workflow is challan confirmation: draft challans do not affect stock, but when a challan is confirmed, the backend validates available stock, prevents negative stock, reduces inventory and creates stock movement audit records inside a database transaction. Challan items store product snapshots so historical sales data stays correct even if product details change later.

8. Final Self-Check Before Submission

- Can you login with all roles?
- Can Sales add customer notes and follow-up dates?
- Can Warehouse see product stock and movement logs?
- Does draft challan avoid stock reduction?
- Does confirmed challan reduce stock exactly once?
- Does insufficient stock return a readable error?
- Does old challan keep old product price after product price changes?
- Are frontend/backend/database live URLs working?
- Does README include setup, env vars, test credentials and limitations?